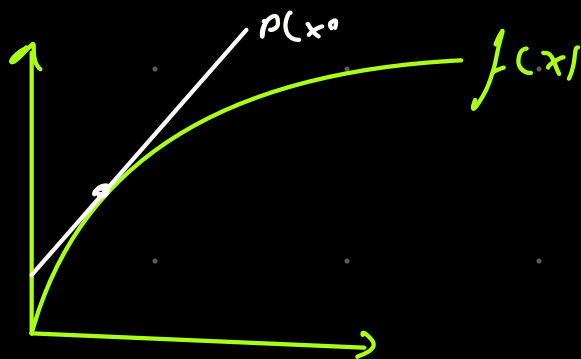


MATERIAS INTRODUÇÃO

Polinômio de Taylor (1D)

$$f(x) \approx p(x) \quad x=x_0$$

aproximar a função a um polinômio em um ponto



$$p_2(x)$$

$$f(x) \approx a + bx + cx^2$$

$$x_0 = 0$$

$$f(0) = p(0) = a$$

$$f'(0) = p'(0) = b$$

$$f''(0) = p''(0) = 2c$$

$$a = f(0)$$

$$b = f'(0)$$

$$c = \frac{f''(0)}{2}$$

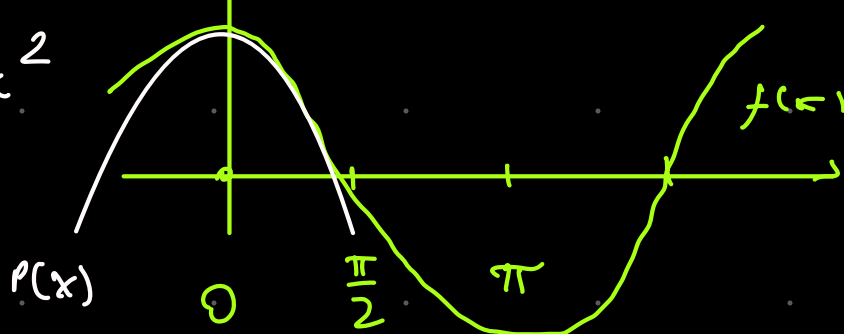
$$p_2(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2$$

Polinômio de Taylor de ordem 2 em torno de $x=0$

Exemplo: $f(x) = \cos x$

$$\cos x \approx \cos(0) - \sin(0)x - \frac{\cos(0)}{2}x^2$$

$$\cos x \approx 1 - \frac{1}{2} x^2$$



$$f(x) \approx f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + \frac{1}{3!} f'''(x_0)(x-x_0)^3$$

$$f(x) \approx \sum_{n=0}^{\infty} \frac{1}{n!} \frac{d^n f}{dx^n} (x-x_0)^n$$

Matrix Diagonalization

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad \text{find a vector basis } \{\vec{u}_1, \vec{u}_2\} \text{ s. that}$$

if a matrix is
Symmetric it is
Diagonalizable

$$\begin{aligned} A \vec{u}_1 &= \lambda_1 \vec{u}_1 \\ A \vec{u}_2 &= \lambda_2 \vec{u}_2 \end{aligned}$$

eigenvalue eigenvector

$$\vec{u} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix}$$

$$\begin{pmatrix} x-y \\ -x+y \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix}$$

$$\begin{pmatrix} x - y - \lambda x \\ -x + y - \lambda y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} (1-\lambda)x - y \\ (1-\lambda)y - x \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} (1-\lambda)x - y = 0 \\ (1-\lambda)y - x = 0 \end{cases} \quad x, y, \lambda$$

Let's imagine we already know a $\vec{u}_1$

If $\vec{u}_1$ is eigenvector $\rightarrow 5\vec{u}_1$ is also an eigenvector
with the same eigenvalue

So for one λ there is an Infinite amount
of eigenvectors S.P.I.

$$\begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} = 0 \quad \begin{aligned} (1-\lambda)^2 &= 1 \\ 1-\lambda &= \pm 1 \end{aligned}$$

$$\begin{aligned} \lambda_1 &= 0 \\ \lambda_2 &= 2 \end{aligned}$$

So for $\lambda_1 = 0$

the one rule it has to follow

$$\begin{cases} x - y = 0 \\ -x + y = 0 \end{cases} \rightarrow \underline{x = y}$$

$$\vec{\mu}_1 = K \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

lets use $\begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$

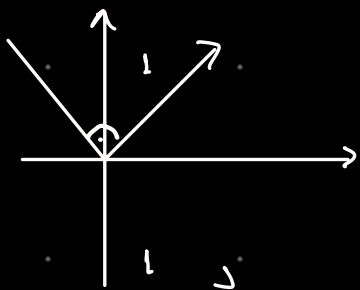
$$\underline{|\vec{\mu}_1| = 1}$$

& for $\lambda_2 = 2$

$$\begin{cases} -x - y = 0 \\ -x - y = 0 \end{cases}$$

$x = -y$ so $\vec{\mu}_2 = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ +\frac{\sqrt{2}}{2} \end{pmatrix}$

Now they form an orthonormal basis



$$\left| \begin{array}{c} \vec{\mu}_1 \\ \vec{\mu}_2 \end{array} \right| > 0 \quad \frac{1}{2} + \frac{1}{2} = 1 > 0$$

$$M = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{cases} \det(M) = 1 \\ \vec{\mu}_1 \cdot \vec{\mu}_2 = \mu_1^T \mu_2 = 0 \end{cases} \Rightarrow \text{they are orthogonal}$$

orthogonal matrix $\rightarrow$ rotational matrix

$$M^{-1} = M^T$$

$$MM^{-1} = I$$

$$\underline{MM^T = I}$$

Test 1

A symmetric $\Leftrightarrow$ orthogonally diagonalizable

$$D = M^T A M = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$$\begin{pmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \end{pmatrix} (A) \begin{pmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$A = M D M^T$$

D is the eigenvalues of A

Example $(x \ y) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$$(x \ y) \begin{pmatrix} x-y \\ -x+y \end{pmatrix} = x^2 - xy - yx + y^2$$

$$x^2 + y^2 - 2xy$$

With a change of variable

$$x^2 + y^2 - 2xy \rightarrow u^2 + v^2$$

$$\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} = u \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + v \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{\sqrt{2}} \left(\begin{pmatrix} u \\ u \end{pmatrix} + \begin{pmatrix} -v \\ v \end{pmatrix} \right)$$

$$x = \frac{u-v}{\sqrt{2}} \quad y = \frac{u+v}{\sqrt{2}}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

$$\begin{aligned}
 x^2 + y^2 - 2xy &= \left(\frac{u-v}{\sqrt{2}} \right)^2 + \left(\frac{u+v}{\sqrt{2}} \right)^2 - 2 \frac{(u-v)(u+v)}{2} \\
 &= \frac{u^2 + v^2 - 2uv + u^2 + v^2 + 2uv - 2(u^2 - v^2)}{2} \\
 &= 2v^2 = \lambda_1 u^2 + \lambda_2 v^2
 \end{aligned}$$

Integrals in GFT

$$F'(x) = f(x) \quad \int f(x) dx = F(x)$$

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

Changing variable

$$\int (2x+1)^7 dx \quad \begin{cases} y = 2x+1 \\ dy = 2dx \end{cases} \quad dx = \frac{dy}{2}$$

$$\int y^7 \frac{dy}{2} = \frac{y^8}{8} = \frac{(2x+1)^8}{16}$$

Double Integrals

$$\int dx \int dy (x + yx^2) = \int dx (xy + \frac{y^2}{2} x^2)$$

Variable Change in Double Integrals

$$\int dx \int dy (x^2 + y^2)$$

$$x^2 + y^2 = c^2$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

Jacobian

$$x = r \cos \theta \quad y = r \sin \theta$$

Jacobian Matrix

$$\frac{\partial (x, y)}{\partial (r, \theta)} = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$

$$\det(J) = r \cos^2 \theta + r \sin^2 \theta = r \quad (x^2 + y^2) = r^2$$

$$\int d\theta \int dr \left| \frac{\partial (x, y)}{\partial (r, \theta)} \right| r^2 \quad \text{substituted}$$

$$\int d\theta \int dr r^3 = \int d\theta \frac{r^4}{4} = \frac{r^4}{4} \theta + C$$

Gaussian Integrals

$$A = \int_{-\infty}^{\infty} e^{-x^2} dx \rightarrow \text{can't be done analytically}$$

Think of another Integral

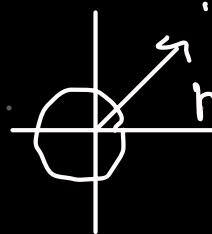
$$A^2 = \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy e^{-x^2-y^2} = \int_{-\infty}^{\infty} dx e^{-x^2} \int_{-\infty}^{\infty} dy e^{-y^2} = \left[\int_{-\infty}^{\infty} dx e^{-x^2} \right]^2$$

$$x = r \cos \theta$$

$$|J| = r$$

$$y = r \sin \theta$$

$$A^2 = \int_0^{2\pi} d\theta \int_0^{\infty} dr \underbrace{r e^{-r^2}}$$



$$A^2 = -\frac{1}{2} \int_0^{2\pi} d\theta \int_0^{\infty} dr -2r e^{-r^2}$$

$$-\frac{1}{2} \int_0^{2\pi} \underbrace{\left[e^{-r^2} \right]_0^{\infty}}_{-1} d\theta$$

$$A^2 = \pi$$

$$A = \sqrt{\pi}$$

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} \quad \begin{aligned} -ax^2 &= -y^2 & \sqrt{a}x &= y \\ \sqrt{a}dx &= dy \end{aligned}$$

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{a}} e^{-y^2} dy = \sqrt{\frac{\pi}{a}}$$

$$f(a) = \int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

Experiment

$$\frac{df(a)}{da} = \frac{d}{da} \int_{-\infty}^{\infty} e^{-ax^2} dx = \int_{-\infty}^{\infty} dx \frac{d}{da} e^{-ax^2} = \frac{d}{da} \sqrt{\frac{\pi}{a}}$$

$$-\frac{\sqrt{\pi}}{2} (a)^{-\frac{3}{2}} = \int_{-\infty}^{\infty} e^{-ax^2} (-x^2) dx$$

$$\int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = \frac{\sqrt{\pi}}{2} a^{-\frac{3}{2}}$$

$$x \rightarrow \phi \quad a \rightarrow m^2$$

$$\int_{-\infty}^{\infty} e^{-\frac{m^2}{2}\phi^2} d\phi = \frac{\sqrt{2\pi}}{m}$$

$$\langle \phi^p \rangle = \frac{1}{m^p} (p-1)(p-3)\dots (5, 3, 1)$$

"functional" progenitor

$$Z[J] = \int_{-\infty}^{\infty} d\phi \exp\left(-\frac{m^2}{2}\phi^2 + J\phi\right)$$

$$\langle \phi \rangle = \frac{Z'[0]}{Z[0]} \quad \langle \phi^2 \rangle = \frac{Z''[0]}{Z[0]}$$

$$\langle \phi^p \rangle = \frac{Z^{(p)}[0]}{Z[0]}$$